S&P GLOBAL HACKATHON · FINANCIAL RISK

Autonomous AI/NLP Financial Risk Engine

From live financial news to explainable portfolio decisions for S&P Global and institutional risk teams.

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THE CASE FOR EARLY SIGNALS

Financial risk appears in text before it appears in prices

Two outputs, one signal layer

TRADITIONAL SYSTEMS

Information latency

Traditional risk systems lean on backward-looking market and reporting data, creating information latency.

Tactical index rebalancing

Dynamic allocation changes from standardized risk signals.

OUR APPROACH

Text becomes signal

Our approach ingests high-velocity financial and geopolitical text, converts it into standardized risk signals, and routes those signals into quantitative decisions.

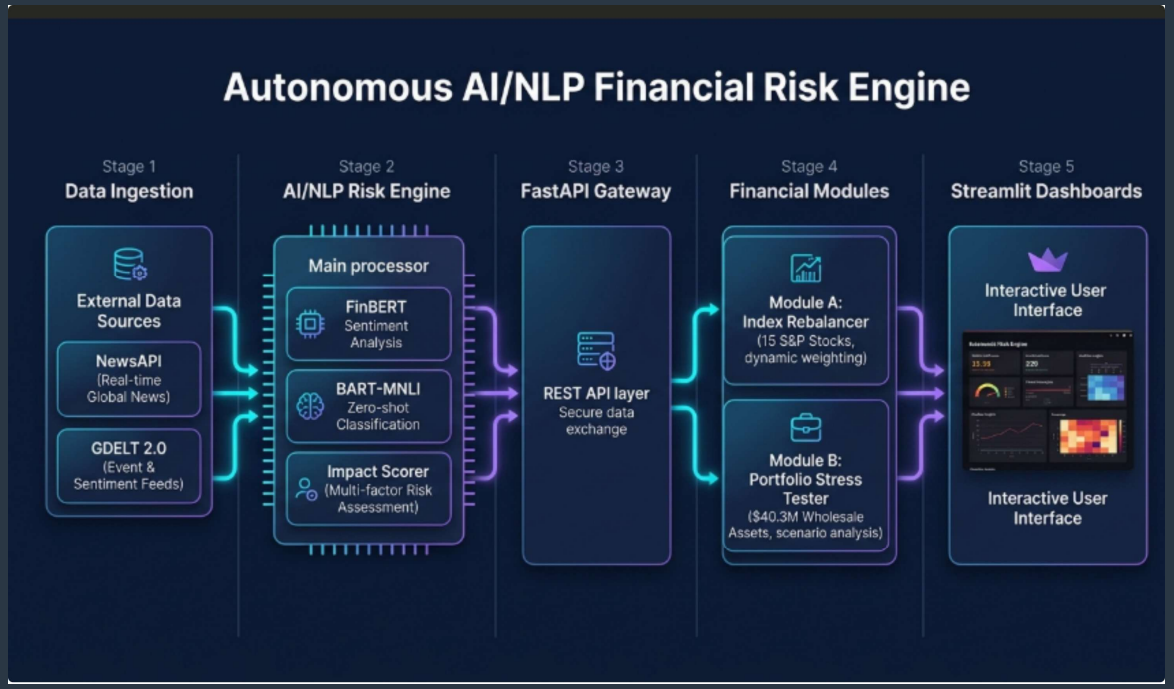
Strategic wholesale portfolio stress testing

Cross-asset scenario analysis when risk becomes systemic.

SYSTEM DESIGN

Five layers turn headlines into decisions

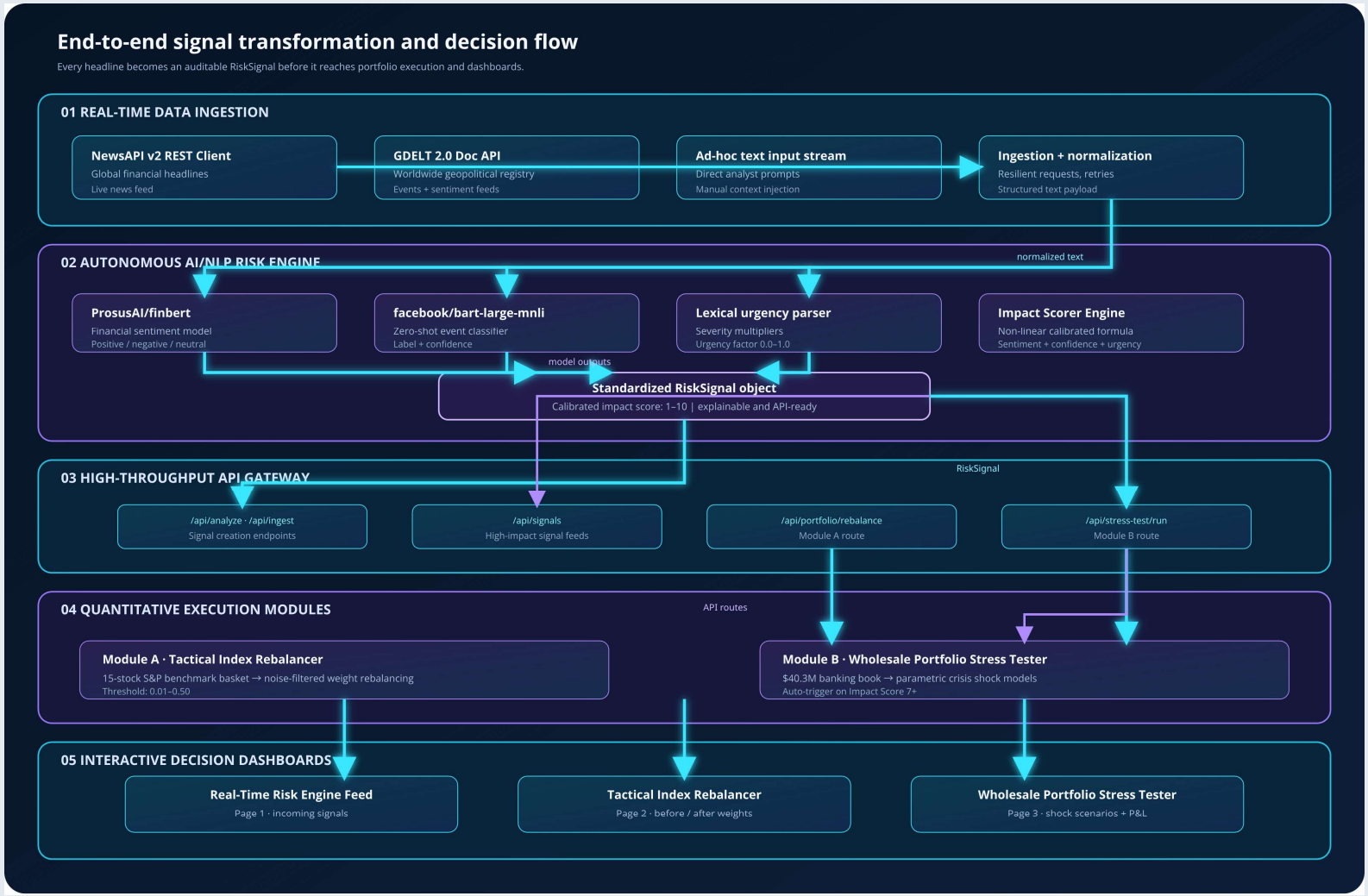
- Ingest live text from NewsAPI, GDELT 2.0, and analysts.
- Use FinBERT and BART-MNLI to classify events and score impact.
- Route risk signals through FastAPI into portfolio tools and dashboards.



Supplied system flow: ingestion → AI/NLP risk engine → FastAPI → financial modules → Streamlit dashboards.

SYSTEM DESIGN

End-to-end pipeline: from text to decisions



IMPLEMENTATION HIGHLIGHTS

Implementation combines transparent formulas with pretrained models

IMPACT SCORE

1.0 + 2.5×sentiment magnitude
+ 3.0×confidence
+ 1.5×urgency
+ systemic-risk bonus

capped at 10

Sentiment

FinBERT produces Positive minus Negative probability, a score from -1.0 to +1.0.

Event taxonomy

BART-MNLI classifies seven categories, including macroeconomic, geopolitical, credit, regulatory, earnings, M&A, and natural disaster events.

Stack

Transformers/PyTorch, FastAPI/Pydantic, NumPy/Pandas, Streamlit/Plotly, and Pytest.

KEY RESULTS

The prototype produces actionable risk signals, not just sentiment labels

Output	Trigger or control	Decision produced
RiskSignal	Sub-second triage of unstructured financial headlines; sentiment, event, confidence, urgency.	Immutable RiskSignal records; Impact score 1–10.
Module A	Filters low-conviction noise with a configurable noise threshold 0.01–0.50.	Rebalances a representative 15-stock S&P blue-chip basket; dynamic 15-stock weights.
Module B	Systemic signals at Impact Score ≥7/10 automatically trigger stress testing across six parametric shock scenarios.	Stress-tests a $40.3M multi-asset wholesale banking portfolio; multi-asset shock PnL. Automated test suite: 15/15 tests passing.

DOMAIN IMPACT

From early warning to capital preservation

Information arbitrage

Detect emerging risk at the headline stage, before it is fully reflected in prices.

Institutional resilience

Project cross-asset drawdowns and identify exposures that need hedging or review.

Explainability

Every portfolio action traces back to model outputs and a visible severity formula.

Regulatory readiness

Accelerate scenario analysis and support CCAR/Basel III-style capital adequacy workflows.

PRODUCTION VALIDATION

Next step: validate the signal-to-decision loop in production-like conditions

Calibrate scores against historical market reactions and labeled financial events.

Add entity linking, deduplication, multilingual coverage, and model monitoring for distribution drift.

Connect to governed S&P Global data products and authenticated enterprise feeds.

Backtest rebalancing and stress scenarios with transaction costs, liquidity constraints, and analyst review.

Current prototype assumptions include synthetic portfolios, representative basket weights, and headline-level semantic density.